

INTRODUCTION TO FIREWALLAD

TASKS

1. Install firewalld on your Linux system, start the service, and verify its status using systemctl and firewall-cmd commands.

Install Firewalld and Verify Status

Step 1: Check if firewalld is installed

dnf list installed firewalld

dnf install firewalld -y

```
[root@localhost ~]# dnf install firewalld -y
Last metadata expiration check: 3:55:47 ago on Tue 14 Jul 2026 11:21:50 AM IST.
Package firewalld-1.3.4-18.el9.noarch is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]# dnf list installed firewalld
Installed Packages
firewalld.noarch                               1.3.4-18.el9                                @anaconda
[root@localhost ~]#
```

Step 2: Start firewalld

systemctl start firewalld

Step 3: Enable at boot

systemctl enable firewalld

Step 4: Verify status

systemctl status firewalld

```
[root@localhost ~]# systemctl start firewalld
[root@localhost ~]# systemctl enable firewalld
[root@localhost ~]# systemctl status firewalld
● firewalld.service - firewalld - dynamic firewall daemon
   Loaded: loaded (/usr/lib/systemd/system/firewalld.service; enabled; preset: enabled)
   Active: active (running) since Tue 2026-07-14 15:09:31 IST; 10min ago
     Docs: man:firewalld(1)
  Main PID: 3449 (firewalld)
    Tasks: 2 (limit: 4272)
   Memory: 19.2M (peak: 36.6M)
      CPU: 1.124s
   CGroup: /system.slice/firewalld.service
           └─3449 /usr/bin/python3 -s /usr/sbin/firewalld --nofork --nopid

Jul 14 15:09:29 localhost.localdomain systemd[1]: Starting firewalld - dynamic firewall daemon>
Jul 14 15:09:31 localhost.localdomain systemd[1]: Started firewalld - dynamic firewall daemon.
lines 1-13/13 (END)
```

Step 5: Verify using firewall-cmd

firewall-cmd --state

```
[root@localhost ~]# firewall-cmd --state
running
[root@localhost ~]#
```

2. Add a new rule to firewalld to allow incoming HTTP (port 80) traffic, then use firewall-cmd to confirm the rule is active.
Hint: Use the **--permanent** and **--reload** options to make changes persistent.

Allow HTTP (Port 80)

Step 1: Add HTTP service

firewall-cmd --permanent --add-service=http

```
[root@localhost ~]# firewall-cmd --permanent --add-service=http
Warning: ALREADY_ENABLED: http
success
[root@localhost ~]#
```

Step 2: Reload firewall

firewall-cmd --reload

```
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]#
```

Step 3: Verify

firewall-cmd --list-services

```
success
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client http ssh
[root@localhost ~]#
```

3. Temporarily block all incoming traffic on port 8080 using firewalld, then test by attempting to access a simple web server (e.g., Python's http.server) running on that port.

Block Port 8080

Step 1: Block port 8080

firewall-cmd --permanent --remove-port=8080/tcp

```
[root@localhost ~]# firewall-cmd --permanent --remove-port=8080/tcp
Warning: NOT_ENABLED: 8080:tcp
success
[root@localhost ~]#
```

Step 2: Reload

firewall-cmd --reload

Step 3: Verify

firewall-cmd --list-ports

```
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]# firewall-cmd --list-ports
[root@localhost ~]#
```

4. List all currently active firewalld zones and display the services and ports allowed in the public zone.

Display Zones and Services

Step 1: List active zones

firewall-cmd --get-active-zones

```
[root@localhost ~]# firewall-cmd --get-active-zones
public
  interfaces: ens160
[root@localhost ~]#
```

Step 2:

Display public zone

firewall-cmd --zone=public --list-all

```
[root@localhost ~]# firewall-cmd --zone=public --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client http ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@localhost ~]#
```

5.Remove a custom rule you added earlier (such as the HTTP port 80 rule) and verify that the change has taken effect using firewall-cmd.

Remove HTTP Rule

Step 1: Remove HTTP

firewall-cmd --permanent --remove-service=http

```
[root@localhost ~]# firewall-cmd --permanent --remove-service=http
success
[root@localhost ~]#
```

Step 2: Reload

firewall-cmd --reload

```
success
[root@localhost ~]# firewall-cmd --reload
success
[root@localhost ~]#
```

Step 3: Verify

firewall-cmd --list-services

```
[root@localhost ~]# firewall-cmd --list-services
cockpit dhcpv6-client ssh
[root@localhost ~]#
```